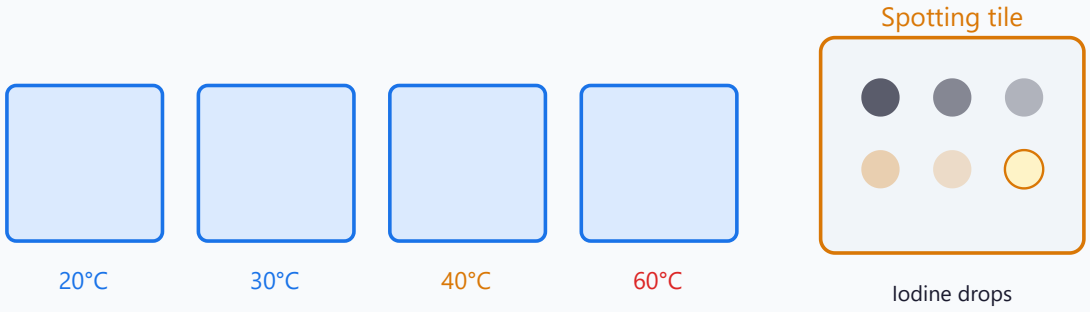
RAG Learning

Edexcel B A-Level Biology — Core Practicals

Table of Contents

1. CP1: Effect of Temperature on Enzyme Activity (Amylase)
2. CP2: Cell Fractionation
3. CP3: Osmosis in Plant Tissue
4. CP4: Effect of Caffeine on Daphnia Heart Rate
5. CP5: Membrane Permeability (Beetroot)
6. CP6: Plant Mineral Deficiencies
7. CP7: Rate of Water Loss from a Leaf
8. CP8: Antimicrobial Properties of Plants
9. CP9: Photosynthesis Rate (Elodea)
10. CP10: Respiration Rate (Respirometer)
11. CP11: Effect of Exercise on Ventilation
12. CP12: Plant Responses to Stimuli (Tropisms)
13. CP13: Genetic Variation in a Population
14. CP14: Biodiversity Using Sampling Techniques
15. CP15: DNA Extraction from a Plant
16. CP16: Effect of Environmental Factor on Species Distribution

CP1: Effect of Temperature on Enzyme Activity (Amylase)



Amylase + starch → sample into iodine every 30 s

End-point: iodine stays yellow-brown (no starch remaining)

Rate = 1 / time to end-point

Apparatus

- Amylase solution, starch solution (1%)
- Iodine solution, white spotting tile
- Water baths at 20, 30, 40, 50, 60 °C
- Test tubes, pipettes, stopwatch, thermometer

Method

1. Place amylase and starch in separate test tubes in a water bath. Equilibrate for 5 minutes.
2. Mix amylase and starch. Start the stopwatch immediately.
3. Every 30 seconds, remove a drop and place on a spotting tile with iodine solution.
4. Record the time at which iodine no longer turns blue-black (all starch digested).
5. Repeat at each temperature (3 repeats). Calculate rate = 1/time.

Safety

- Iodine is an irritant and stains skin — wear gloves.
- Care with hot water baths — use test tube holders.

Analysis

- Plot rate (1/time) against temperature. Expect a bell-shaped curve.
- Optimum temperature for amylase is around 37 °C. Above this, the enzyme denatures.
- Below the optimum, rate increases as kinetic energy increases (more successful collisions between enzyme and substrate).

Sources of Error

Error	Effect	Improvement
Tubes not equilibrated	Reaction starts at wrong temperature	Equilibrate both tubes for 5 minutes
Sampling interval too long	Miss the exact end-point	Sample every 15–30 seconds near end-point
Judging colour change subjectively	Inconsistent end-point	Use same person for all judgements; use colorimeter
Contamination of iodine wells	False readings	Use fresh iodine drop for each sample

Variables

Variable Type	Variable	How Controlled
Independent	Temperature	Set by water baths (use thermometer to verify)
Dependent	Time for starch digestion (then rate = 1/t)	Timed with stopwatch; iodine test for end-point
Controlled	Volume and concentration of amylase and starch	Use same pipette and stock solutions
Controlled	pH	Use buffer solution (pH 7)

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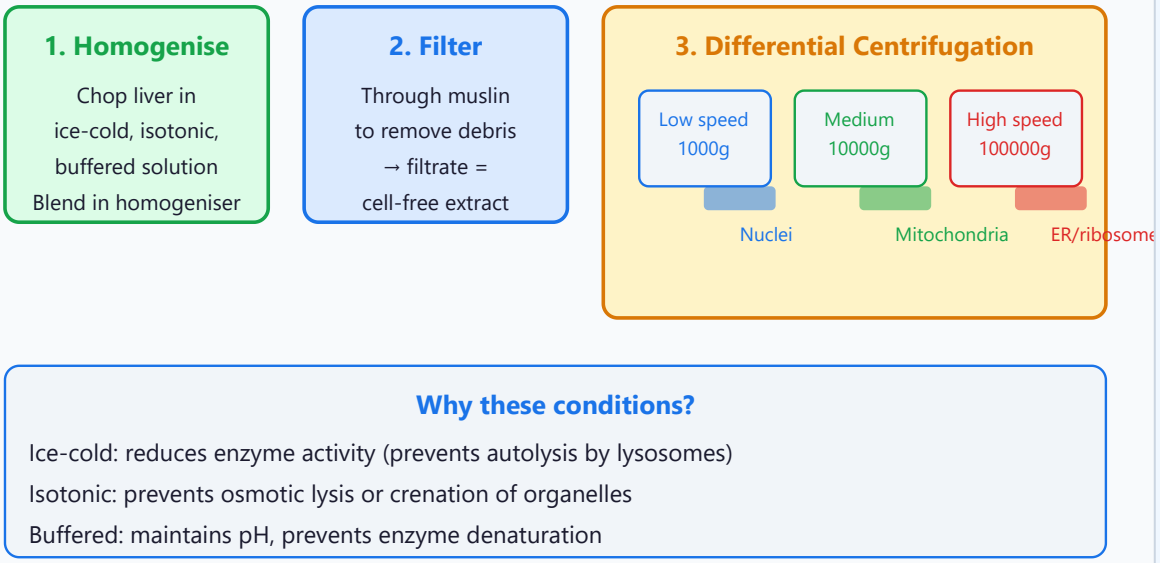
Controlled	Equilibration time	5 minutes for both solutions before mixing

Expected Results

Rate increases with temperature up to the optimum (~37 °C for salivary amylase) because higher kinetic energy means more frequent and energetic collisions between enzyme and substrate, increasing the likelihood of forming enzyme-substrate complexes. Above the optimum, hydrogen and ionic bonds holding the tertiary structure break, the active site changes shape (denaturation), and the substrate can no longer bind. At very high temperatures (60+ °C), the enzyme is completely denatured and the rate drops to zero. The Q_{10} value (the factor by which rate increases per 10 °C rise) is typically around 2 for enzyme reactions below the optimum.

Exam Tip: Calculate rate as 1/t (where t is the time for starch to be completely digested). This gives the initial rate of reaction. Explain the decrease in rate above the optimum in terms of *denaturation* — the active site changes shape permanently and can no longer form enzyme-substrate complexes. Never say the enzyme is "killed" (it is a protein, not alive) or that it is "destroyed" (the molecule still exists, but its shape has changed irreversibly).

🔬 CP2: Cell Fractionation



Why these conditions?

Ice-cold: reduces enzyme activity (prevents autolysis by lysosomes)
Isotonic: prevents osmotic lysis or crenation of organelles
Buffered: maintains pH, prevents enzyme denaturation

Apparatus

- Fresh liver or cauliflower tissue
- Ice-cold isotonic buffered solution (e.g. cold sucrose solution, 0.25 M)
- Homogeniser (blender), muslin cloth, beakers
- Centrifuge, centrifuge tubes, ice

Method

1. Chop tissue finely. Add to ice-cold, isotonic, buffered sucrose solution.
2. Homogenise to break open cells (release organelles into solution).
3. Filter through muslin to remove cell debris and connective tissue.
4. Centrifuge filtrate at low speed (1000 g, 10 min). Pellet = nuclei. Decant supernatant.
5. Centrifuge supernatant at medium speed (10,000 g, 30 min). Pellet = mitochondria.
6. Centrifuge at high speed (100,000 g, 60 min). Pellet = ER and ribosomes.

Safety

- Wear gloves when handling animal tissue.
- Centrifuge must be balanced to prevent damage or breakage.

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- Keep solutions ice-cold throughout — handle with care.

Analysis

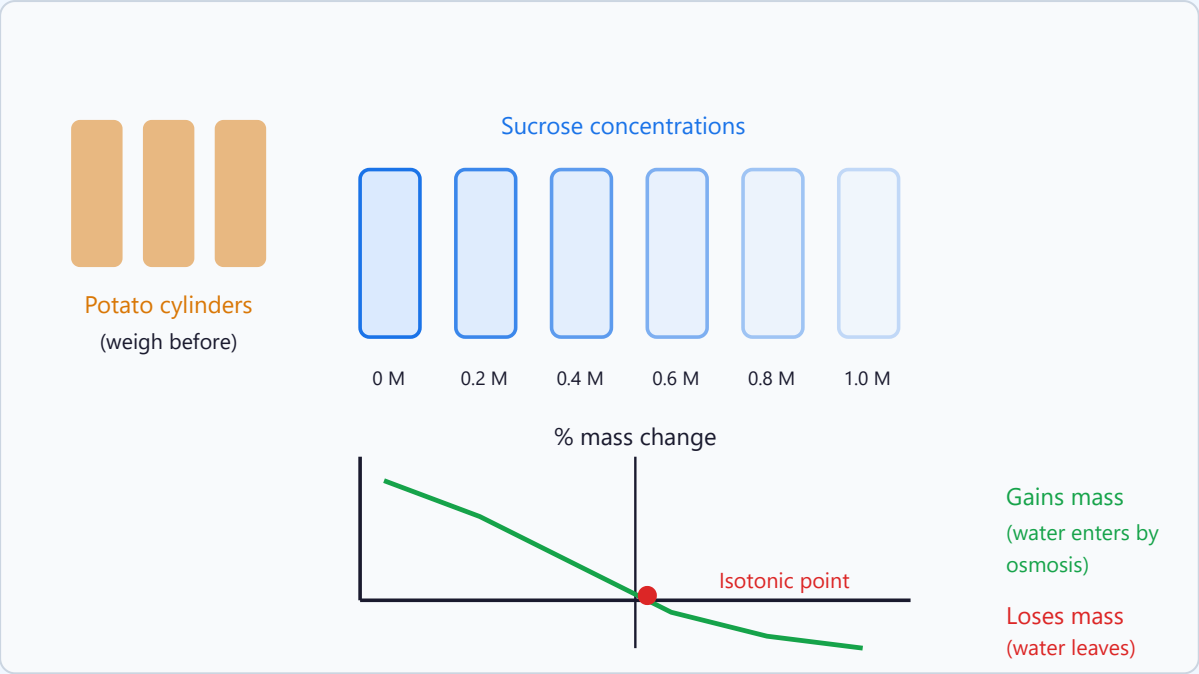
- Organelles separate by size and density: heaviest (nuclei) sediment first at lowest speed.
- Each pellet can be resuspended and tested (e.g. for enzyme activity).
- Confirm identity using staining or biochemical assays.

Sources of Error

Error	Effect	Improvement
Solution not ice-cold	Enzymes digest organelles	Keep on ice throughout; work quickly
Solution not isotonic	Organelles burst or shrink	Use 0.25 M sucrose solution
Centrifuge not balanced	Uneven separation; equipment damage	Always balance tubes in centrifuge
Incomplete homogenisation	Not all cells broken open	Homogenise thoroughly; check with microscope

Exam Tip: Know the three conditions for the solution: *ice-cold* (prevents autolysis by lysosomes), *isotonic* (prevents osmotic damage), *buffered* (maintains pH). Organelles sediment in order of decreasing size and mass.

🔬 CP3: Osmosis in Plant Tissue



Apparatus

- Potato, cork borer, scalpel, ruler
- Sucrose solutions at 0, 0.2, 0.4, 0.6, 0.8, 1.0 mol dm^{-3}
- Test tubes or boiling tubes, balance (to 0.01 g)
- Paper towels, labels, stopwatch

Method

1. Cut potato cylinders of equal length using a cork borer. Blot dry and weigh each one.
2. Place one cylinder in each sucrose concentration. Leave for 30–60 minutes.
3. Remove, blot dry gently, reweigh.
4. Calculate % change in mass: (final − initial) / initial × 100.
5. Repeat 3 times per concentration.

Safety

- Care with sharp cutting tools — cut on a white tile.
- Sucrose solutions are non-hazardous.

Analysis

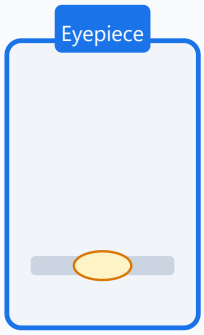
- Plot % change in mass (y-axis) against sucrose concentration (x-axis).
- Where the line crosses zero = isotonic point (water potential of potato cells equals external solution).
- Below this concentration: cells gain water (hypotonic solution). Above: cells lose water (hypertonic solution).

Sources of Error

Error	Effect	Improvement
Cylinders different sizes	Different surface areas affect rate	Use same cork borer; trim to equal length
Inconsistent blotting	Variable surface water affects mass	Blot once each side; same technique
Not enough time left in solution	Equilibrium not reached	Leave for at least 30 minutes
Potato from different parts of tuber	Different water potential	Use cylinders from same area of same potato

Exam Tip: The isotonic point (where % mass change = 0) gives the solute potential of the potato cells. Water moves from high water potential to low water potential by osmosis. Always use *percentage* change in mass, not absolute change, so different starting masses are accounted for.

🔬 CP4: Effect of Caffeine on Daphnia Heart Rate



Caffeine concentrations: 0%, 0.1%, 0.3%, 0.5%

Fresh Daphnia for each concentration

Count beats in 20 s × 3 = beats per minute

Allow 2 min for caffeine to take effect before counting

Ethics: use low concentrations, return to pond water, minimise stress

Apparatus

- Daphnia, cavity slides, cotton wool strands, pipette
- Microscope (low power), caffeine solutions (0–0.5%)
- Pond water, stopwatch

Method

1. Place Daphnia on cavity slide in pond water. Use cotton wool to restrict movement.
2. Observe heart under low-power microscope. Count beats for 20 s × 3 at rest. Calculate mean bpm.
3. Add caffeine solution. Wait 2 minutes. Recount beats (3 × 20 s).
4. Use a fresh Daphnia for each concentration. Return all to pond water.

Safety

- Ethical treatment: minimise time on slide, return to habitat.
- Use low-power LED to avoid heating.

Analysis

- Plot mean heart rate (bpm) against caffeine concentration.
- Caffeine inhibits phosphodiesterase, preventing breakdown of cAMP, leading to increased heart rate.

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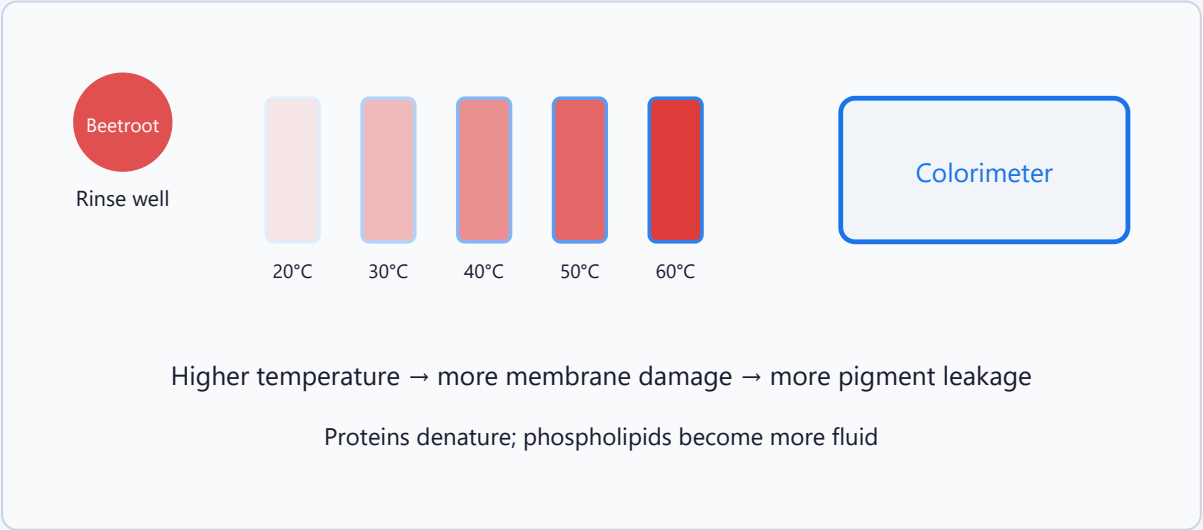
Sources of Error

Error	Effect	Improvement
Heart too fast to count accurately	Underestimate rate	Tap dots on paper in time with beats; count dots
Heat from lamp	Heart rate increases independently	Use LED light source
Cumulative caffeine effect	Inaccurate readings	Use fresh Daphnia for each concentration

Exam Tip: Always discuss ethical considerations: use minimum numbers of organisms, low concentrations, allow recovery, return to habitat. Explain why a fresh Daphnia is used for each concentration.

CP5: Membrane Permeability (Beetroot)



Apparatus

- Fresh beetroot, cork borer, scalpel, white tile

- Test tubes, distilled water, water baths (20–70 °C)
- Colorimeter and cuvettes, thermometer

Method

1. Cut uniform beetroot discs. Rinse thoroughly to remove surface pigment.
2. Place one disc in each tube with 5 cm^3 distilled water. Incubate at each temperature for 20 min.
3. Remove beetroot. Measure absorbance using colorimeter.
4. Repeat 3 times per temperature. Calculate means.

Safety

- Sharp tools — cut on white tile away from body.
- Hot water — use tongs for tubes.

Analysis

- Plot mean absorbance against temperature.
- Absorbance increases sharply above ~40 °C as membrane proteins denature and phospholipid bilayer loses integrity.

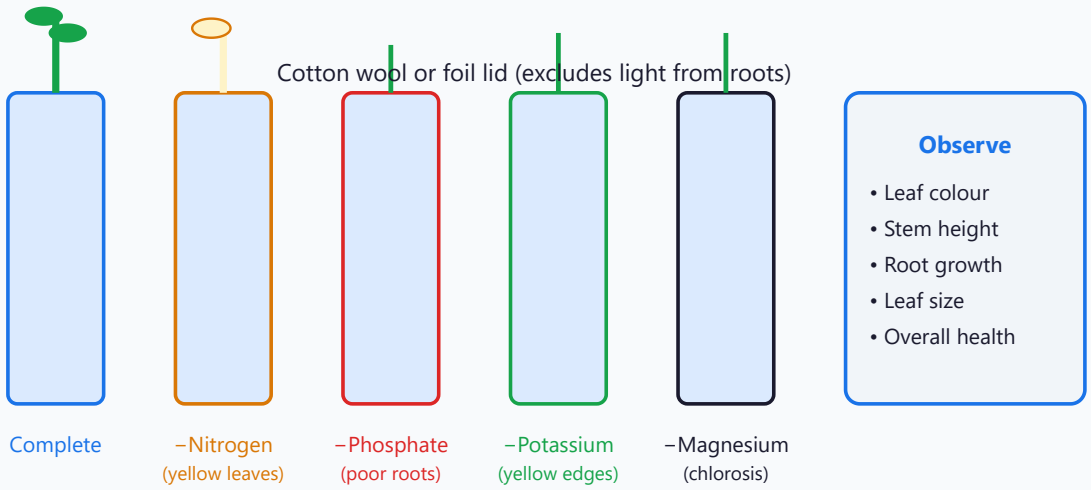
Sources of Error

Error	Effect	Improvement
Unequal disc sizes	Different surface area	Use same cork borer and trim uniformly
Insufficient rinsing	Surface pigment inflates readings	Rinse until water is clear
Fingerprints on cuvette	Inaccurate absorbance	Handle cuvette by top; wipe with tissue

Exam Tip: Explain results using the fluid mosaic model: membrane proteins denature (tertiary structure disrupted by heat), creating gaps. Phospholipids gain kinetic energy and the bilayer

becomes more permeable, allowing anthocyanin to diffuse out.

CP6: Plant Mineral Deficiencies



Grow for 2–4 weeks in mineral solutions lacking one ion each

Apparatus

- Seedlings (e.g. wheat or lettuce), boiling tubes or large test tubes
- Complete mineral solution and solutions each lacking one ion (N, P, K, Mg, etc.)
- Cotton wool or aluminium foil lids, labels
- Light source, ruler, camera

Method

1. Place seedlings in boiling tubes containing mineral solutions. Support with cotton wool at the top.
2. Set up: complete solution (control), minus nitrogen, minus phosphate, minus potassium, minus magnesium.
3. Place all in the same light and temperature conditions.

4. Observe and record symptoms (leaf colour, growth rate, root development) every few days for 2–4 weeks.

Safety

- Mineral solutions are generally low hazard. Wear eye protection.
- Wash hands after handling solutions and plant material.

Analysis

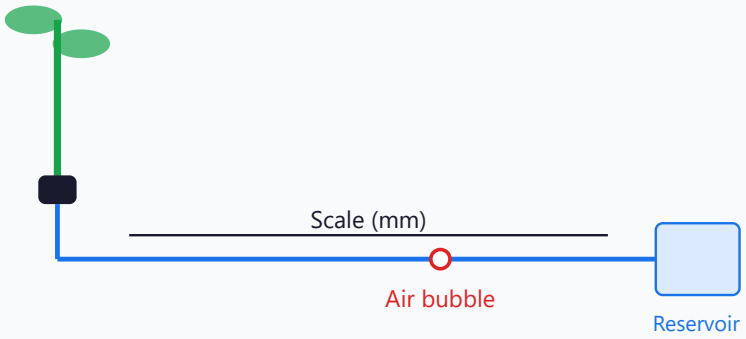
- Without nitrogen: stunted growth, yellowing of older leaves (nitrogen is mobile).
- Without phosphate: poor root growth, purple/dark leaves.
- Without potassium: yellow leaf edges, poor fruit development.
- Without magnesium: chlorosis (yellow leaves with green veins) as Mg is part of chlorophyll.

Sources of Error

Error	Effect	Improvement
Contamination of solutions	Minerals present when should be absent	Use deionised water; clean apparatus
Variable seedling size	Different starting points	Select seedlings of similar size
Algal growth in solutions	Uses up minerals; blocks light	Cover tubes with foil; change solutions weekly

Exam Tip: Know the role of each mineral ion: nitrogen (amino acids and proteins), phosphate (DNA and ATP), potassium (enzyme cofactor, stomatal opening), magnesium (chlorophyll). Deficiency symptoms reflect the role of the missing ion.

🔬 CP7: Rate of Water Loss from a Leaf



Vary: wind (fan), humidity (bag), light, temperature

Rate = distance moved by bubble / time

Apparatus

- Potometer, leafy shoot, rubber tubing
- Capillary tube with scale, water reservoir
- Fan, lamp, plastic bag, Vaseline
- Stopwatch, ruler

Method

1. Cut shoot underwater. Assemble potometer underwater to prevent air locks.
2. Seal all joints with Vaseline. Allow 5 min to equilibrate.
3. Introduce an air bubble. Record distance moved per minute.
4. Test under different conditions: windy (fan on), humid (plastic bag over leaves), bright light, darkness.
5. Reset bubble between tests using reservoir. Repeat 3 times per condition.

Safety

- Sharp tools for cutting stems. Keep electrical equipment away from water.

Analysis

- Rate of water uptake ≈ rate of transpiration.

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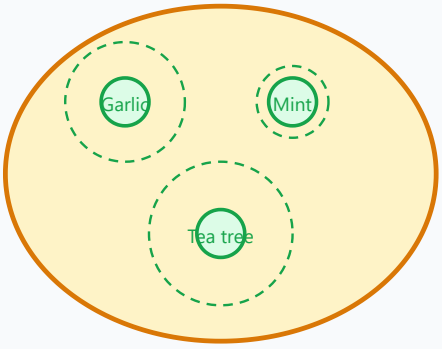
- Higher wind speed, lower humidity, higher temperature, and brighter light all increase transpiration rate.

Sources of Error

Error	Effect	Improvement
Air lock in xylem	No water uptake; bubble doesn't move	Cut stem underwater
Leaks at joints	Water loss not through leaves	Vaseline all connections
Wilting	Stomata close, reducing transpiration	Keep in water; do not delay setup

Exam Tip: A potometer measures water *uptake*, not transpiration directly. Transpiration creates a transpiration pull (cohesion-tension theory), drawing water up the xylem. State how each environmental factor affects stomatal opening and the water potential gradient.

CP8: Antimicrobial Properties of Plants



Bacterial lawn on agar

Method

- Soak discs in extract
- Place on bacterial lawn
- Incubate ≤25°C
- Measure zone πr^2

Apparatus

- Nutrient agar plates, bacterial broth culture
- Plant extracts (garlic, mint, tea tree, etc.), filter paper discs
- Sterile forceps, Bunsen burner, marker pen, ruler

Method

1. Prepare plant extracts by crushing in a pestle and mortar with water or ethanol.
2. Soak sterile filter paper discs in each extract.
3. Spread bacteria on agar plate using aseptic technique.
4. Place soaked discs on the agar with flamed forceps. Include a control disc (water/ethanol only).
5. Seal, label, incubate upside down at ≤25 °C for 24–48 hours.
6. Measure zones of inhibition. Calculate area using πr^2.

Safety

- Do not open plates after incubation. Autoclave before disposal.
- Aseptic technique throughout. Incubate ≤25 °C.

Analysis

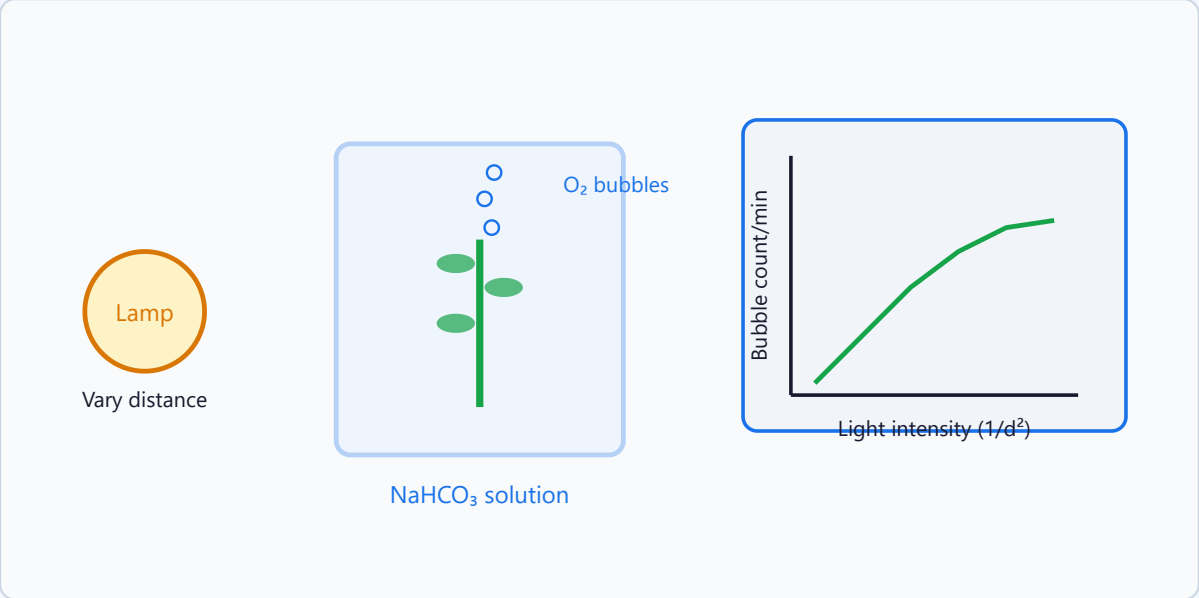
- Compare areas of inhibition zones for different plant extracts.
- Larger area = more effective antimicrobial plant extract.
- The control disc shows whether the solvent alone has any antimicrobial effect.

Sources of Error

Error	Effect	Improvement
Variable extract concentrations	Unfair comparison between plants	Use same volume and concentration method
Discs not equally soaked	Different amounts of extract per disc	Soak for same time; use same disc size
Contamination	Extra colonies obscure zones	Strict aseptic technique

Exam Tip: Plants produce antimicrobial chemicals as a defence against pathogens. Always include a control disc with just the solvent to check it is the plant extract, not the solvent, that inhibits bacterial growth.

CP9: Photosynthesis Rate (Elodea)



Apparatus

- Elodea (pondweed), beaker, $NaHCO_3$ solution
- Bench lamp, ruler, stopwatch
- Thermometer, heat shield (beaker of water)

Method

1. Place Elodea in $NaHCO_3$ solution. Cut stem at an angle. Wait 2 min to equilibrate.
2. Position lamp at measured distance. Count bubbles per minute (3 repeats).
3. Move lamp to different distances (5, 10, 15, 20, 25 cm). Repeat counts.
4. Light intensity $\propto 1/d^2$.

Safety

- Keep lamp away from water (electrical hazard).
- Use heat shield between lamp and beaker.

Analysis

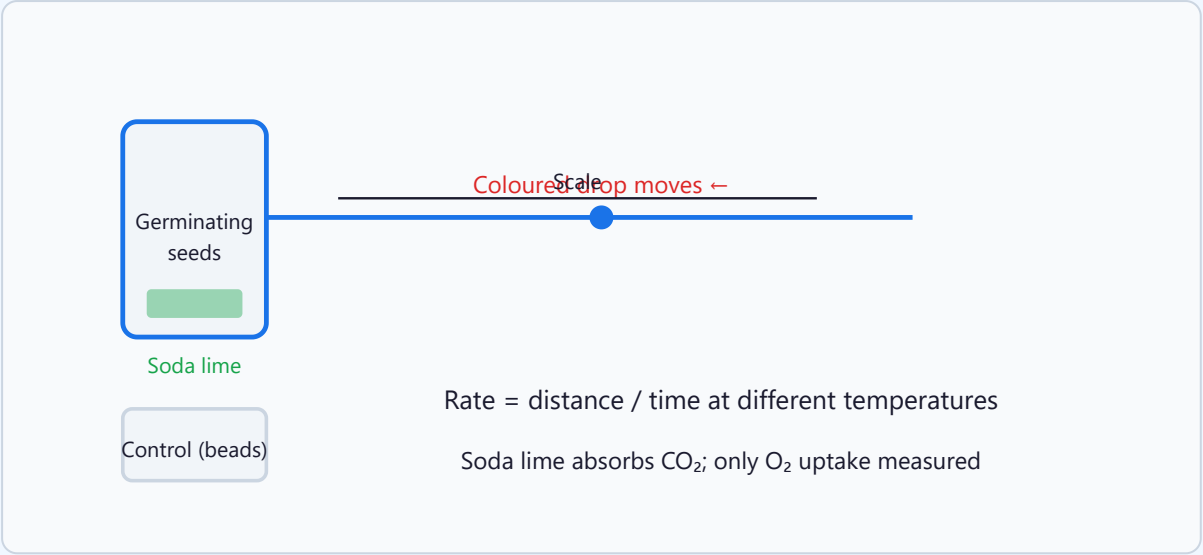
- Plot bubble count per minute against $1/d^2$ (light intensity).
- Rate increases then plateaus (another factor becomes limiting: CO_2 or temperature).

Sources of Error

Error	Effect	Improvement
Unequal bubble sizes	Volume of O_2 varies per bubble	Use gas syringe for volume measurement
Heat from lamp	Temperature confounds results	Use heat shield; monitor temperature
Background light	Extra photosynthesis at all distances	Darken room; use blackout

Exam Tip: $NaHCO_3$ ensures CO_2 is not limiting. Light intensity = $1/d^2$. The plateau indicates another factor has become limiting. Explain limiting factors using a graph with clear annotations.

🔬 CP10: Respiration Rate (Respirometer)



Apparatus

- Simple respirometer, germinating seeds, glass beads (control)
- Soda lime, cotton wool, capillary tube with scale
- Coloured fluid, water bath, stopwatch

Method

1. Set up respirometer with seeds and soda lime. Control uses glass beads.
2. Seal, place in water bath, equilibrate 5 min.
3. Introduce coloured fluid drop. Record distance moved per minute for 10 min.
4. Repeat at different temperatures.

Safety

- Soda lime is corrosive — gloves and eye protection.

Analysis

- Rate = distance / time. Plot rate vs temperature.
- Q_{20} effect: rate roughly doubles per 10 °C (up to optimum).

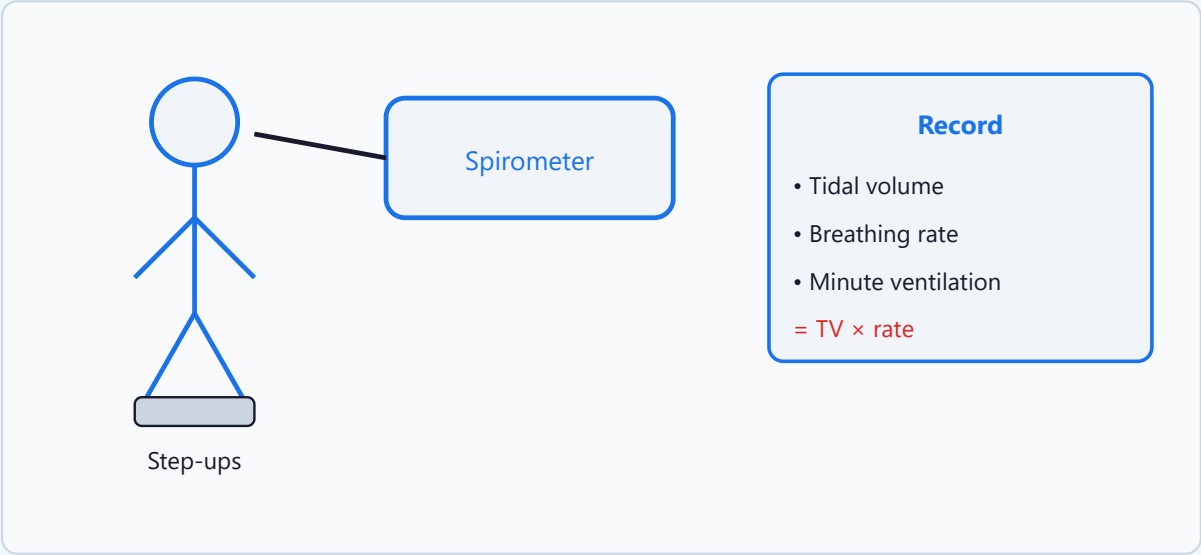
Sources of Error

Error	Effect	Improvement

Leaks	Fluid doesn't move	Seal with Vaseline
No equilibration	Temperature not correct at start	Wait 5 min before recording
No control	Cannot account for pressure changes	Always include glass bead control

Exam Tip: The control tube with glass beads accounts for changes in atmospheric pressure and temperature that are not due to respiration. Soda lime absorbs CO_2, so only O_2 consumption causes volume change.

CP11: Effect of Exercise on Ventilation



Apparatus

- Spirometer or peak flow meter, clean mouthpieces, nose clip
- Step or exercise bike, stopwatch, metronome

Method

1. Measure resting breathing rate and tidal volume.
2. Exercise (e.g. step-ups for 5 min at set rate with metronome).

3. Immediately measure breathing rate and tidal volume. Repeat during recovery every minute.
4. Calculate minute ventilation = tidal volume × breathing rate.

Safety

- Screen for health conditions. Stop if unwell. Use individual mouthpieces.

Analysis

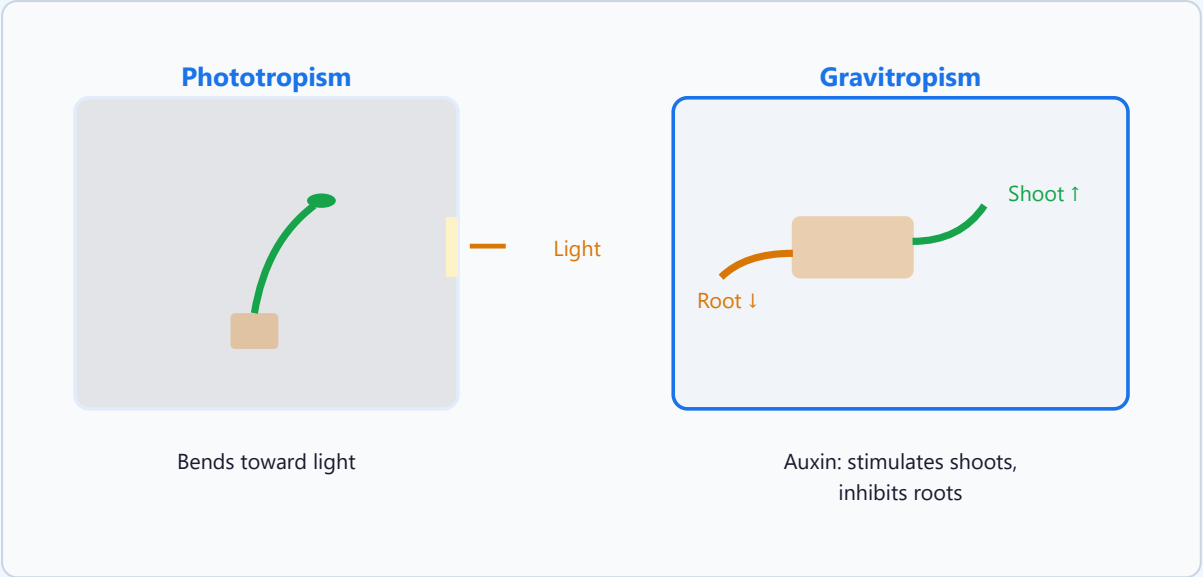
- Both tidal volume and breathing rate increase during exercise.
- CO_2 increase detected by chemoreceptors in medulla and carotid bodies.

Sources of Error

Error	Effect	Improvement
Variable exercise intensity	Inconsistent results	Use metronome; standardise step height and rate
Nose not clipped	Air escapes through nose	Use nose clip during spirometry

Exam Tip: Minute ventilation = tidal volume × breathing rate. During exercise, increased CO_2 is detected by chemoreceptors, which send impulses to the ventilation centre in the medulla oblongata, increasing the rate and depth of breathing.

🛸 CP12: Plant Responses to Stimuli (Tropisms)



Apparatus

- Seedlings (cress or oat coleoptiles), pots or Petri dishes
- Cardboard box with hole, clinostat, light source
- Ruler, protractor, camera

Method

1. Grow seedlings to ~2 cm tall in identical conditions.
2. **Phototropism:** Place in light-proof box with unidirectional light source. Observe bending over 5–7 days.
3. **Gravitropism:** Place seedling on its side. Observe root and shoot bending over 24–48 hours.
4. Control: seedling on clinostat (eliminates gravity as a variable).
5. Measure angle of curvature with protractor. Take photographs.

Safety

- Wash hands after handling soil. Keep electrical equipment dry.

Analysis

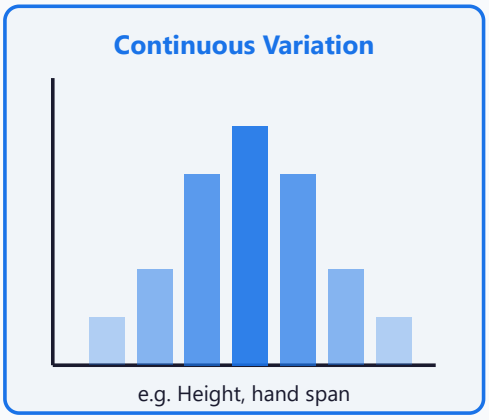
- Auxin moves to shaded side of shoot → greater elongation → bends toward light.
- In roots, auxin accumulates on lower side → inhibits elongation → root grows downward.

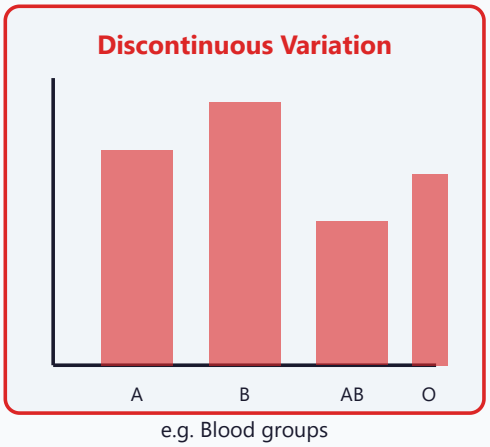
Sources of Error

Error	Effect	Improvement
Light leaks into box	Not truly unidirectional	Seal box thoroughly
Variable seedling ages	Different sensitivities to stimuli	Germinate all seeds at same time
Clinostat speed too fast/slow	Doesn't eliminate gravity properly	Use manufacturer's recommended speed

Exam Tip: Auxin (IAA) causes cell elongation. In shoots, high auxin = more elongation (positive phototropism). In roots, high auxin = less elongation (positive gravitropism). This is because roots and shoots have different sensitivities to auxin concentration.

CP13: Genetic Variation in a Population





Calculate: mean, standard deviation, range. Test: t-test or chi-squared

Apparatus

- Tape measure, ruler, calipers
- Recording sheets, calculator
- Sample of at least 30 individuals (classmates or organisms)

Method

1. Choose a characteristic to measure: continuous (height, hand span, leaf length) or discontinuous (tongue rolling, earlobe attachment, blood group).
2. Measure the chosen characteristic in at least 30 individuals.
3. Record data in a table. Organise into classes/categories.
4. For continuous data: calculate mean, standard deviation, and plot a histogram.
5. For discontinuous data: plot a bar chart. Calculate ratios.

Safety

- Be sensitive when collecting personal measurements.
- Ensure consent from participants.

Analysis

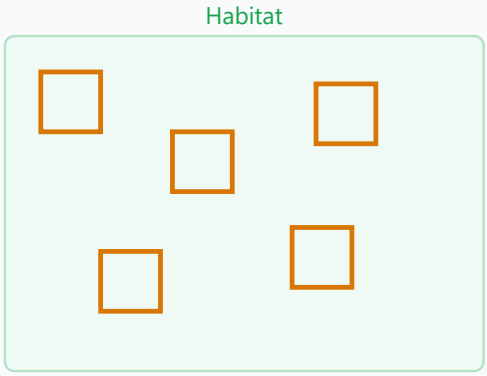
- Continuous variation: normal distribution (bell curve), influenced by polygenic inheritance and environment.
- Discontinuous variation: distinct categories, usually controlled by one or few genes.
- Use standard deviation to quantify the spread of data. Use t-test to compare two populations.

Sources of Error

Error	Effect	Improvement
Small sample size	Does not represent population	Sample at least 30 individuals
Measurement inconsistency	Unreliable data	Same person measures; standardise technique
Non-random sample	Bias in data	Use random selection of individuals

Exam Tip: Continuous variation is polygenic (many genes) and influenced by environment. Discontinuous variation is typically monogenic. Standard deviation measures the spread — a large SD means more variation. A t-test determines if two means are significantly different.

CP14: Biodiversity Using Sampling Techniques



Simpson's Index

$$D = 1 - \Sigma(n/N)^2$$

n = number of one species
N = total number of all species

D close to 1 = high biodiversity
D close to 0 = low biodiversity

Random sampling with quadrats; identify and count all species

Apparatus

- Quadrats (0.5 m × 0.5 m), tape measure
- Random number table, identification keys
- Recording sheets, calculator

Method

1. Use random number coordinates to place quadrats in the habitat.
2. In each quadrat, identify and count every species present.
3. Repeat for at least 10 quadrats. Record totals for each species.
4. Calculate Simpson's Diversity Index: $D = 1 - \Sigma(n/N)^2$.

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Safety

- Fieldwork risk assessment. Watch for hazardous plants/animals.
- Wash hands after handling soil.

Analysis

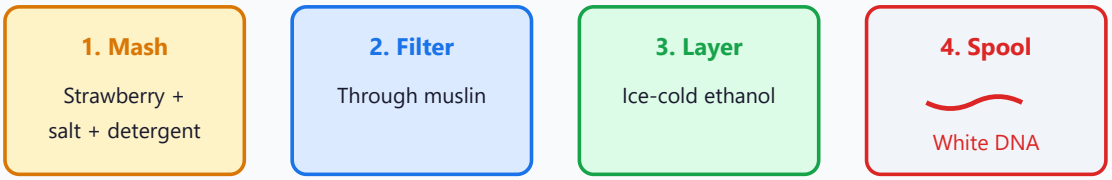
- D close to 1 indicates high biodiversity (many species, even distribution).
- D close to 0 indicates low biodiversity (dominated by one species).
- Compare two habitats by calculating D for each.

Sources of Error

Error	Effect	Improvement
Non-random placement	Sampling bias	Use random number table
Too few quadrats	Unrepresentative	Use at least 10 quadrats
Misidentification	Incorrect species counts	Use identification keys; take photos

Exam Tip: Always show your working for Simpson's index. $D = 1 - \Sigma(n/N)^2$. Calculate (n/N) for each species, square each, sum them, then subtract from 1. Higher D means greater biodiversity. Random sampling avoids bias.

🔬 CP15: DNA Extraction from a Plant



Detergent: dissolves phospholipid membranes (cell + nuclear)
Salt: makes DNA clump (neutralises phosphate charges)
Ice-cold ethanol: DNA is insoluble → precipitates as white threads

Apparatus

- Strawberries, zip-lock bag, extraction buffer (salt + detergent + warm water)
- Muslin, beaker, test tube, ice-cold ethanol (>95%), glass rod

Method

1. Mash strawberries in bag with warm extraction buffer (3 min).
2. Filter through muslin into test tube (one-third full).
3. Gently layer equal volume of ice-cold ethanol on top.
4. Wait 2–3 min. White DNA precipitates at the boundary. Spool with glass rod.

Safety

- Ethanol is flammable — no naked flames.
- Detergent is an irritant — wash hands.

Analysis

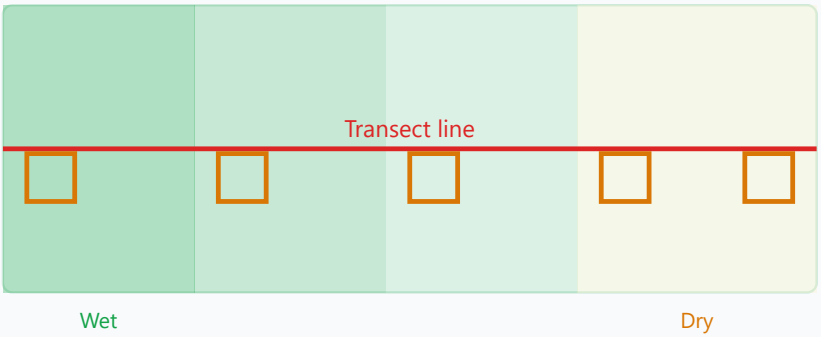
- Detergent lyses membranes, salt causes DNA to aggregate, cold ethanol precipitates DNA.
- Strawberries are octoploid (8n) so yield lots of DNA.

Sources of Error

Error	Effect	Improvement
Ethanol not cold enough	Less DNA precipitates	Keep in freezer before use
Vigorous mixing	DNA shears into small fragments	Layer gently; do not shake
Insufficient mashing	Less DNA released	Mash thoroughly for 3+ minutes

Exam Tip: Know the function of each step: detergent dissolves phospholipid bilayer, salt neutralises DNA's negative phosphate charges so molecules aggregate, ice-cold ethanol causes precipitation because DNA is insoluble in alcohol.

🔬 CP16: Effect of Environmental Factor on Species Distribution



Record

- Species
- % cover
- Soil moisture
- Light
- pH

Spearman's rank correlation: test association between species distribution and abiotic factor

Apparatus

- Tape measure (30 m), quadrats (0.5 m × 0.5 m)
- Identification keys, recording sheets

- Abiotic measurement tools: soil moisture meter, light meter, pH probe, thermometer

Method

1. Lay a transect line across a habitat where an environmental gradient exists (e.g. from stream to drier ground, from tree canopy to open).
2. Place quadrats at regular intervals (e.g. every 2 m) along the transect.
3. At each quadrat: identify species, estimate % cover or count individuals.
4. Measure abiotic factors (soil moisture, light, pH, temperature) at each point.
5. Record all data in a table.

Safety

- Fieldwork risk assessment (terrain, weather, working near water).
- Wash hands after handling soil.

Analysis

- Plot a kite diagram or bar chart showing species distribution along the transect.
- Use Spearman's rank correlation to test for a significant association between an abiotic factor and species distribution.
- If the calculated r_s exceeds the critical value at $p \leq 0.05$, the correlation is significant.

Sources of Error

Error	Effect	Improvement
Too few sampling points	Pattern not detected	Sample at close intervals; many quadrats
Abiotic factors change during day	Measurements not comparable	Take all measurements at same time of day
Misidentification of species	Inaccurate data	Use identification keys; take photos

Trampling affects habitat	Alters results in subsequent quadrats	Walk carefully; avoid stepping on quadrat area

Exam Tip: Spearman's rank is used for correlation between two sets of data where one or both are ordinal or not normally distributed. If r_s is close to +1 or −1, there is a strong positive or negative correlation. Always compare to the critical value at the chosen significance level ($p \leq 0.05$).